

Development of Stratified Stochastic Tensor Contraction

With Applications to Multicomponent Coupled-Cluster
and Green's Function methods

$$\sum_{abcij} \approx \sum_{\mathbf{k}} \text{🎲}$$

Ari Chakraborty

Department of Chemistry

Syracuse University

Acknowledgements



■ Chandler Martin,



Arick Grootveld



Anaira Jalan



Prof. Gandikota
(Dept. of Electrical
Engg. & Computer
Science)

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Development of Stratified Stochastic Tensor Contraction Method for Applications in Electronic Structure Theory

Chandler Martin,¹ Arick Grootveld,² Anaira Jalan,³ Venkata Gandikota,² and Arindam Chakraborty^{4, a)}

¹⁾ *Department of Physics, Syracuse University, Syracuse, New York 13244 USA*

²⁾ *Department of Electrical Engineering and Computer Science, Syracuse University, Syracuse, New York 13244 USA*

³⁾ *Department of Physics, Imperial College London, London SW7 2AZ, United Kingdom*

⁴⁾ *Department of Chemistry, Syracuse University, Syracuse, New York 13244 USA*



Chandler Martin,



Arick Grootveld



Anaira Jalan



Prof. Gandikota

Overview of this talk

▪Objective:

Development of efficient scheme for performing large-scale tensor contraction

▪Motivation:

- Tensor contractions are widespread in the field of quantum chemistry, scientific computing, and numerical linear algebra

- We need an efficient method when the computational effort needed to perform a tensor contraction exceeds our computation budget (in terms of number of FLOPS)

▪Strategy:

- Performing tensor contraction stochastically
- Development of Stratified Stochastic Tensor Contraction (SSTC) method

▪Chemical applications:

- Semiconductor quantum dots: CdSe, CdS, PbSe, PbS
- Metallic nanoparticles: Ag and Au, Au/Cu alloys

Sequential vs Stochastic Tensor Contraction

Tensor Contraction:

$$T = \sum_{i_1 i_2 i_3 i_4 \dots i_8}^N A(i_1, i_2) B(i_3, i_4) C(i_5, i_6) D(i_7, i_8) G(i_1, i_2, \dots, i_8)$$

Serial Tensor Contraction:

- All terms are evaluated sequentially in a multi-nested loop
- Gives exact result

Stochastic Tensor Contraction:

- Not all terms are evaluated
- Evaluation of terms is performed stochastically
- Gives approximate results

SSTC: Stratified Stochastic Tensor Contraction

Stochastic Enumeration Method for Counting NP-Hard Problems

Reuven Rubinstein

Methodol Comput Appl Probab (2013) 15:249–291
DOI 10.1007/s11009-011-9242-y

See also: Diagrammatic Monte Carlo, FCI-QMC

Stratified Stochastic Tensor Contraction

Tensor Contraction:

$$T = \sum_{i_1 i_2 i_3 i_4 \dots i_8}^N A(i_1, i_2) B(i_3, i_4) C(i_5, i_6) D(i_7, i_8) G(i_1, i_2, \dots, i_8)$$

Composite index K:

$$T = \sum_{K=1}^{K_{\max}} \Lambda_K$$

Stratification of the composite index space:

$$K = \{1, \dots, M, \dots, 2M, \dots, 3M, \dots, K_{\max}\}$$

Spanedda, N., McLaughlin, P. F., Beyer, J. J., & Chakraborty, A. (JCTC 2022)

Stratified Stochastic Tensor Contraction

Stratification of the composite index space:

$$K = \{\underbrace{1, \dots, M}_{\text{segment1}}, \underbrace{\dots, 2M}_{\text{segment2}}, \dots, 3M, \dots, \underbrace{K_{\max}}_{\text{segment}D}\}$$

Segmented sum:

$$T = \sum_{K=1}^{K_{\max}} \Lambda_K = \sum_{\alpha=1}^{N_{\text{seg}}} \sum_{j=1}^{M_{\alpha}} \Lambda_{\alpha j} \quad S = S_1 + S_2 + \dots + S_{N_{\text{seg}}}$$

Segmented Average:

$$\mu_{\alpha} = \frac{S_{\alpha}}{M_{\alpha}} \quad S_{\alpha} = M_{\alpha} \mu_{\alpha}$$

Spanedda, N., McLaughlin, P. F., Beyer, J. J., & Chakraborty, A. (JCTC 2022)

Stratified Stochastic Tensor Contraction

Stochastic Evaluation of Segment Average:

$$\bar{X}_\alpha = \frac{1}{N_{\text{segment}}^\alpha} \sum_{\substack{j \in M \\ \text{swr}}}^{N_{\text{segment}}^\alpha} \Lambda_{\alpha j}$$

▪ Estimate segment average by randomly sampling N_{sample} points from that segment **without replacement**

▪ **swr: Sample Without Replacement**

Limiting condition:

$$\lim_{N_{\text{sample}} \rightarrow M_\alpha} \bar{X}_\alpha = \mu_\alpha$$

▪ Stochastic average approaches the exact segment average as the number of sampling points approach segment population

Stratified Stochastic Tensor Contraction

$$S = S_1 + S_2 + \dots + S_{N_{\text{seg}}}$$

Sequential Summation

$$S_\alpha = M_\alpha \mu_\alpha$$

- **Fixed cost:** Fixed number of FLOPS because all terms are included

Stratified Stochastic Summation

$$S_\alpha \approx M_\alpha \bar{X}_\alpha$$

- **Variable cost:** Total number of FLOPS depends on the number of sampling points used

Stratified Stochastic Summation: Potential for optimization

- **Sample less** for segments that contribute less towards the overall result
- **Sample more** for segments that generate high statistical error

SSTC: Analysis of Sampling Error

Exact result:

$$T_{\text{exact}} = \sum_{i_1 i_2 i_3 i_4 \dots i_8}^N A(i_1, i_2) B(i_3, i_4) C(i_5, i_6) D(i_7, i_8) G(i_1, i_2, \dots, i_8)$$

$$T_{\text{exact}} = N_{\text{totpop}} \times \bar{T}_{\text{exact}}$$

SSTC result:

$$T_{\text{SSTC}} = N_{\text{totpop}} \times \bar{T}_{\text{SSTC}}$$

- SSTC provides an estimation of the exact mean of the tensor contraction through sampling
- SSTC is guaranteed to give the exact result if all points in the tensor contraction are sampled

$$\lim_{N_{\text{sample}} \rightarrow N_{\text{totpop}}} \bar{T}_{\text{SSTC}} = \bar{T}_{\text{exact}}$$

SSTC: Analysis of Sampling Error

$$T_{\text{SSTC}} = N_{\text{totpop}} \times \bar{T}_{\text{SSTC}}$$

- The variance of $\bar{T}$ is related to the sampling error: $\mathbb{V}[\bar{T}_{\text{SSTC}}]$

$$\lim_{N_{\text{sample}} \rightarrow N_{\text{totpop}}} \bar{T}_{\text{SSTC}} = \bar{T}_{\text{exact}}$$

$$\lim_{N_{\text{sample}} \rightarrow N_{\text{totpop}}} \mathbb{V}[\bar{T}_{\text{SSTC}}] = 0$$

- The SSTC variance is related to the variance in each segment:

$$\mathbb{V}[\bar{T}_{\text{SSTC}}] = \sum_{h=1}^{N_{\text{seg}}} w_h^2 \mathbb{V}[\bar{x}_h]$$

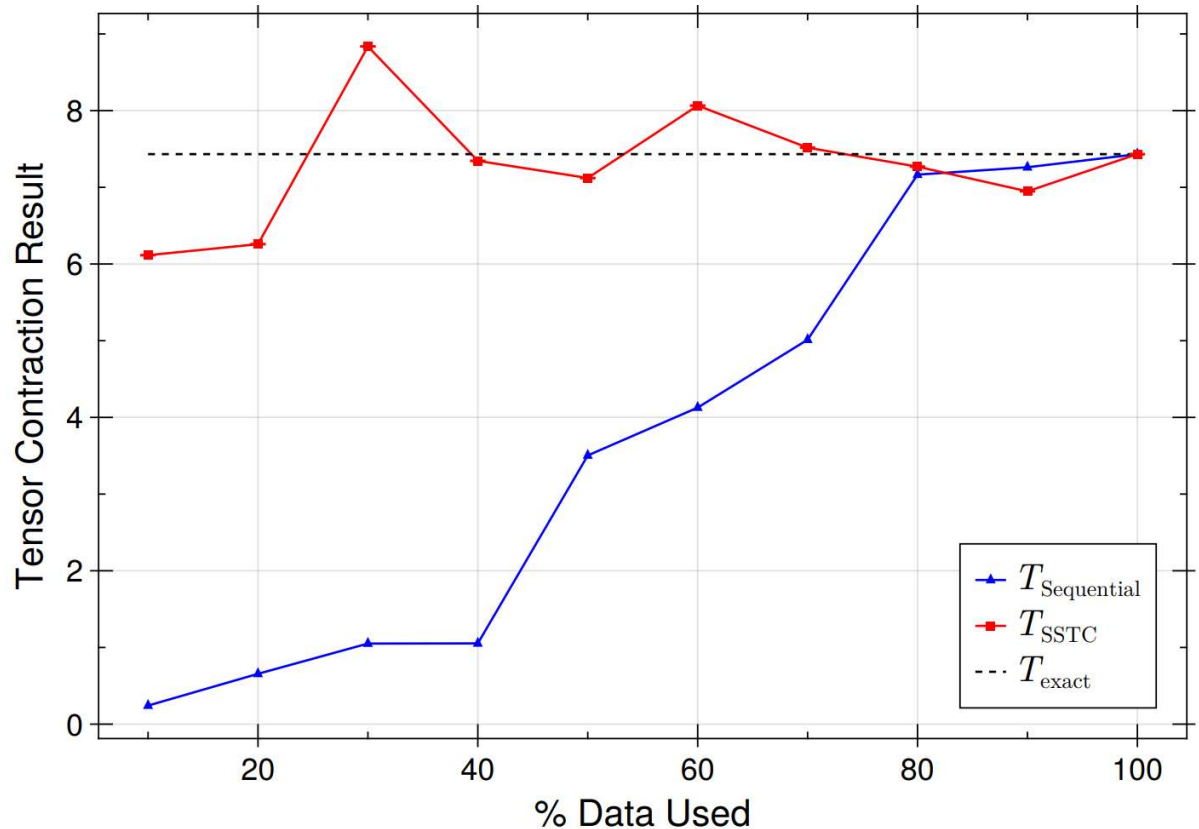
- The weight w_h is the fraction of points in each segment: $w_h = \frac{N_h^{\text{segtot}}}{N_{\text{totpop}}}$

Key Points:

- We can minimize the sampling error by optimal redistribution of sampling points over segments

SSTC: Application to Streamed Data

- The entire tensor is too big and is never constructed or stored
- User has access to only the elements of the tensor: `get_tij(...)`
- Elements of tensor are streamed and access to elements is time-ordered



Key Points:

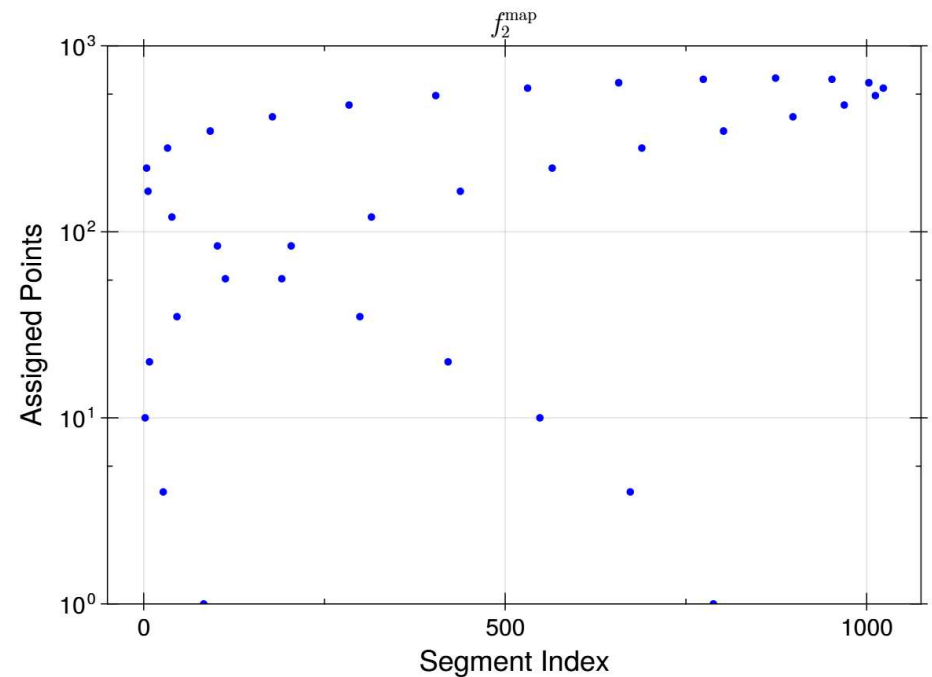
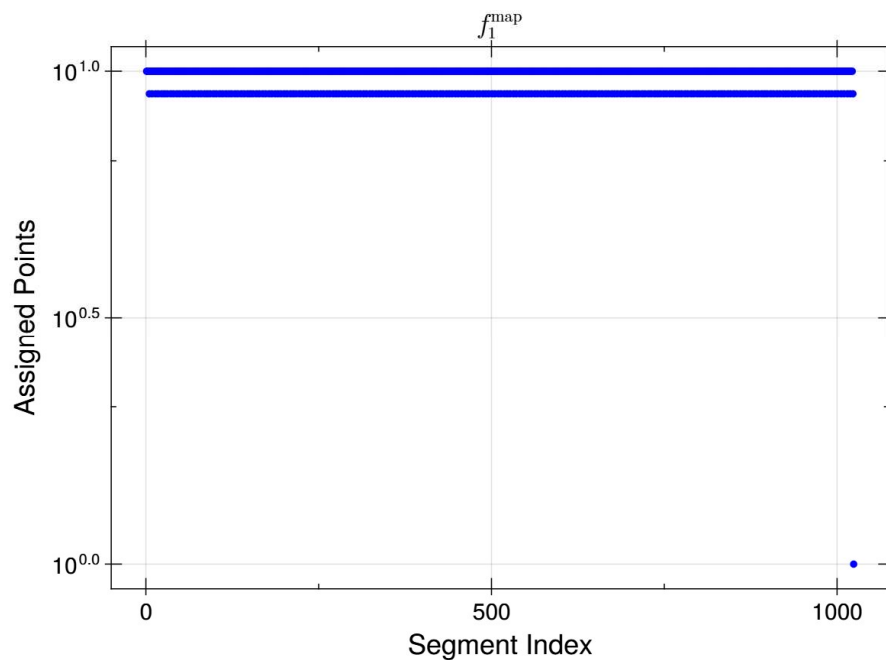
- SSTC is able to provide an estimate of the full tensor contraction result

SSTC: Application to Streamed Data

- The assignment of an incoming data to a specific segment is done using a mapping function

$$A(i_1, i_2)B(i_3, i_4)C(i_5, i_6)D(i_7, i_8)G(i_1, i_2, \dots, i_8) \rightarrow [i_1, \dots, i_D]$$

$$[i_1, \dots, i_D] \xrightarrow{f_{\text{map}}} k_{\text{seg}}$$

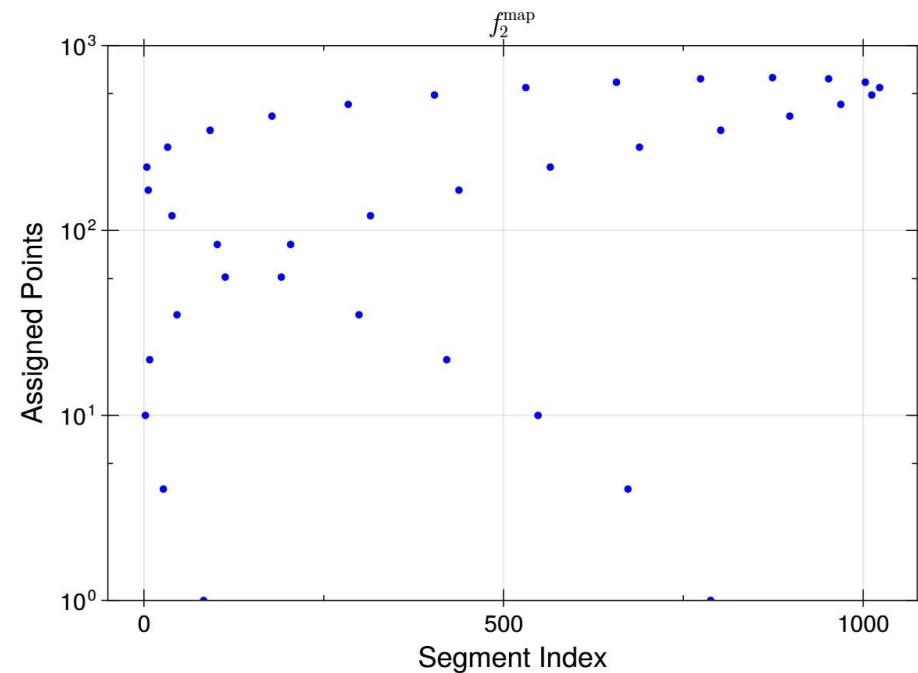
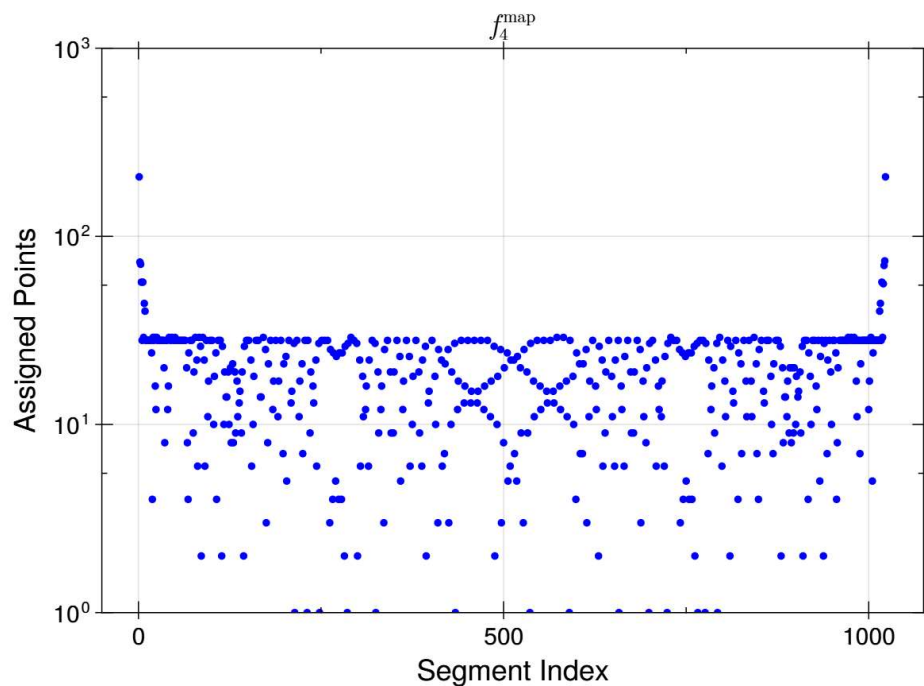


SSTC: Application to Streamed Data

- The assignment of an incoming data to a specific segment is done using a mapping function

$$A(i_1, i_2)B(i_3, i_4)C(i_5, i_6)D(i_7, i_8)G(i_1, i_2, \dots, i_8) \rightarrow [i_1, \dots, i_D]$$

$$[i_1, \dots, i_D] \xrightarrow{f_{\text{map}}} k_{\text{seg}}$$



SSTC: Variance Reduction using Mapping Function

- The assignment of an incoming data to a specific segment is done using a mapping function

$$[i_1, \dots, i_D] \xrightarrow{f_{\text{map}}} k_{\text{seg}}$$

$$\begin{array}{ll} \Lambda(i_1, i_2, \dots, i_D) & \xrightarrow{f_A} k_1 \\ \Lambda(j_1, j_2, \dots, j_D) & \xrightarrow{f_A} k_5 \end{array} \qquad \begin{array}{ll} \Lambda(i_1, i_2, \dots, i_D) & \xrightarrow{f_B} k_{21} \\ \Lambda(j_1, j_2, \dots, j_D) & \xrightarrow{f_B} k_{15} \end{array}$$

- **Keypoint#1:** The mapping functions only depend on the indices and not on the value of the kernel
- **Keypoint#2:** The ideal mapping function will have the property:

$$\begin{array}{ll} \Lambda(i_1, i_2, \dots, i_D) & \xrightarrow{f_{\text{ideal}}} k_{\alpha} \\ \Lambda(j_1, j_2, \dots, j_D) & \xrightarrow{f_{\text{ideal}}} k_{\beta} \end{array} \qquad k_{\alpha} < k_{\beta} \Rightarrow \Lambda(\mathbf{i}) < \Lambda(\mathbf{j}) \quad \text{for any two pair of indices}$$

SSTC: Variance Reduction using Mapping Function

- The ideal mapping function will have the property:

$$\begin{array}{ll} \Lambda(i_1, i_2, \dots, i_D) & \xrightarrow{f_{\text{ideal}}} k_\alpha \\ \Lambda(j_1, j_2, \dots, j_D) & \xrightarrow{f_{\text{ideal}}} k_\beta \end{array} \quad k_\alpha < k_\beta \Rightarrow \Lambda(\mathbf{i}) < \Lambda(\mathbf{j}) \quad \text{for any two pair of indices}$$

- **Why is this important:** The ideal mapping function is able to rank order the kernel without even looking at the kernel!!

- The variance of the SSTC mean is a functional of the mapping function

$$\mathbb{V}[\bar{T}_{\text{SSTC}}] = \sum_{h=1}^{N_{\text{seg}}} w_h^2 [f_{\text{map}}] \mathbb{V}[\bar{x}_h]$$

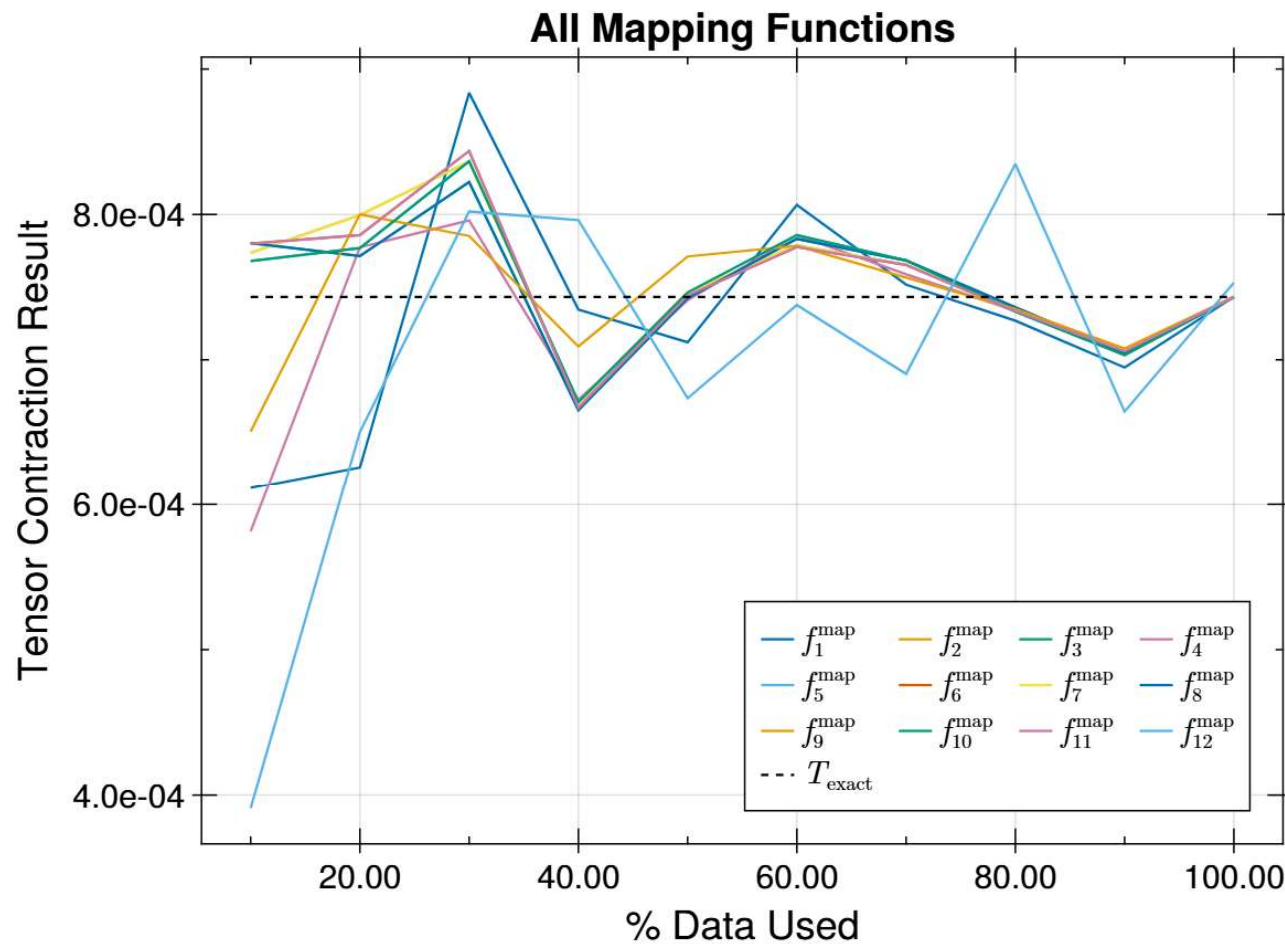
- **Keypoint:**

Sampling error can be systematically reduced through functional minimization

TABLE VI. Mapping functions used for the SSTC method. $i_{\max} = \max([i_1, i_2, \dots, i_D])$ is the maximum value in the index vector. $i_{\min} = \min([i_1, i_2, \dots, i_D])$ is the analogous minimum value.

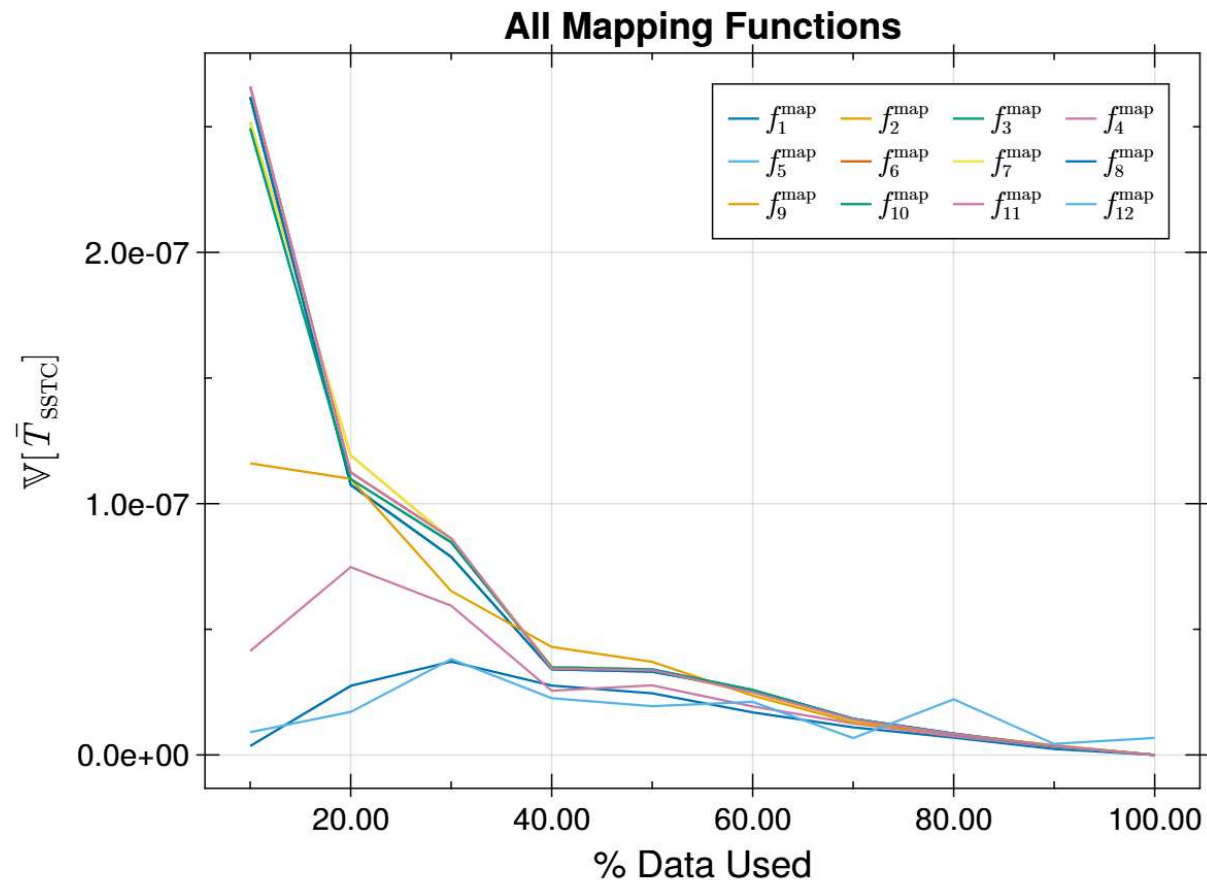
Function 1	$x = -1 + 2 \frac{K - K_{\min}}{K_{\max} - K_{\min}}$
Function 2	$x = \cos(\text{avg}([i_1, i_2, \dots, i_D]))$
Function 3	$x = \cos(i_{\max} - i_{\min})$
Function 4	$x = \cos(K)$
Function 5	$x = \cos(i_{\max})$
Function 6	$x = \cos(i_{\min})$
Function 7	$x = \sin(\text{avg}([i_1, i_2, \dots, i_D]))$
Function 8	$x = \sin(i_{\max} - i_{\min})$
Function 9	$x = \sin(K)$
Function 10	$x = \sin(i_{\max})$
Function 11	$x = \sin(i_{\min})$
Function 12	$x \sim \mathcal{U}(-1, 1)$

- Choose the mapping function that minimizes variance to the average tensor contraction



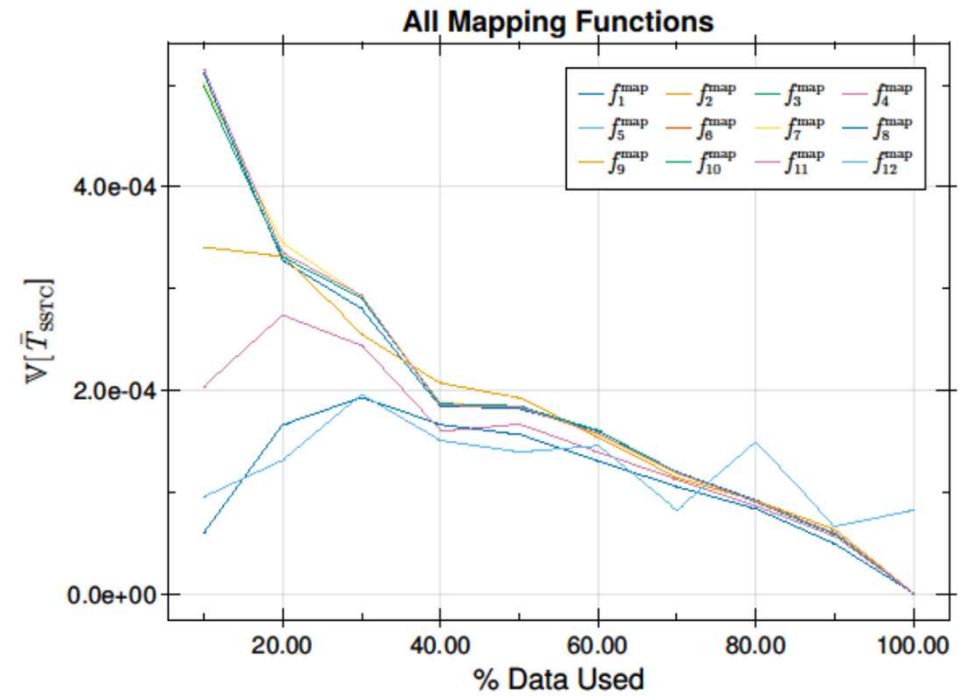
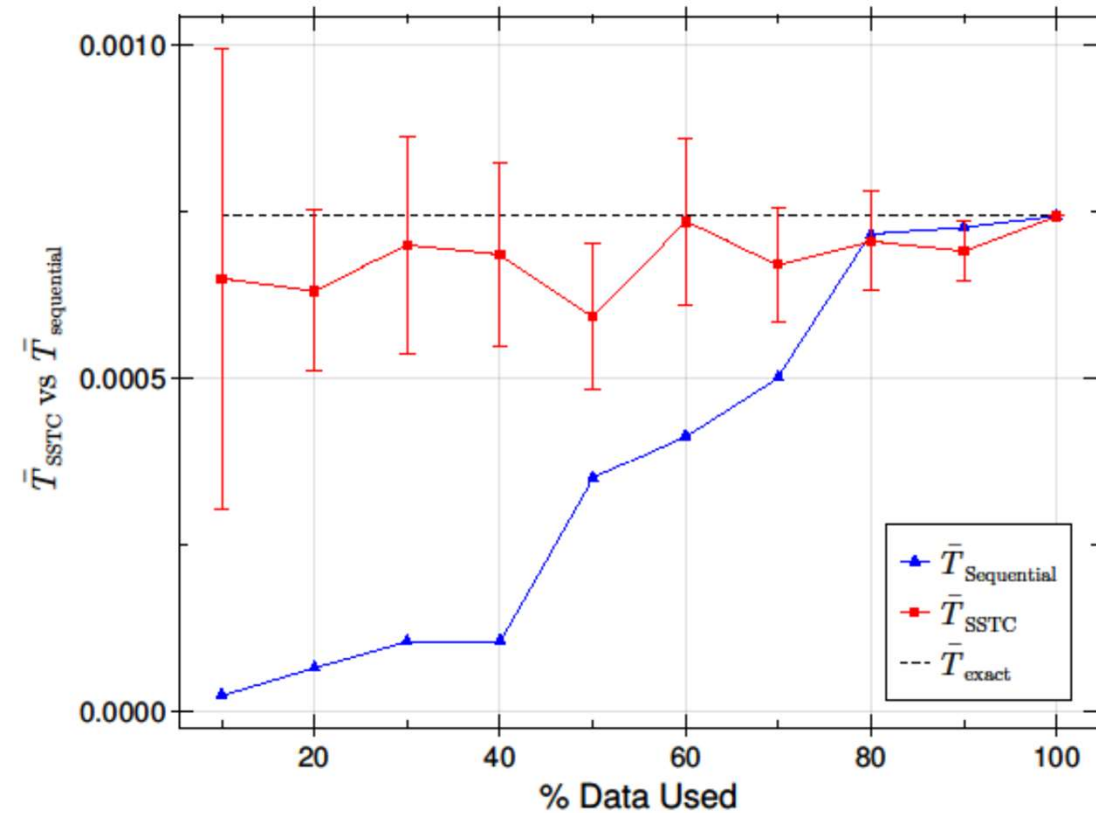
SSTC: Application to Streamed Data

- Choose the mapping function that minimizes variance to the average tensor contraction



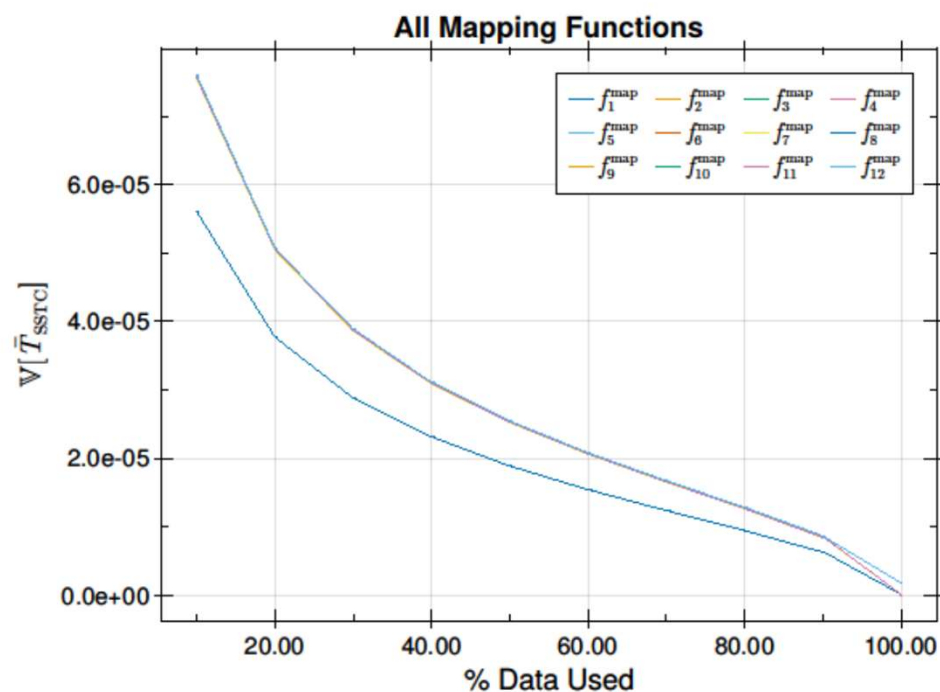
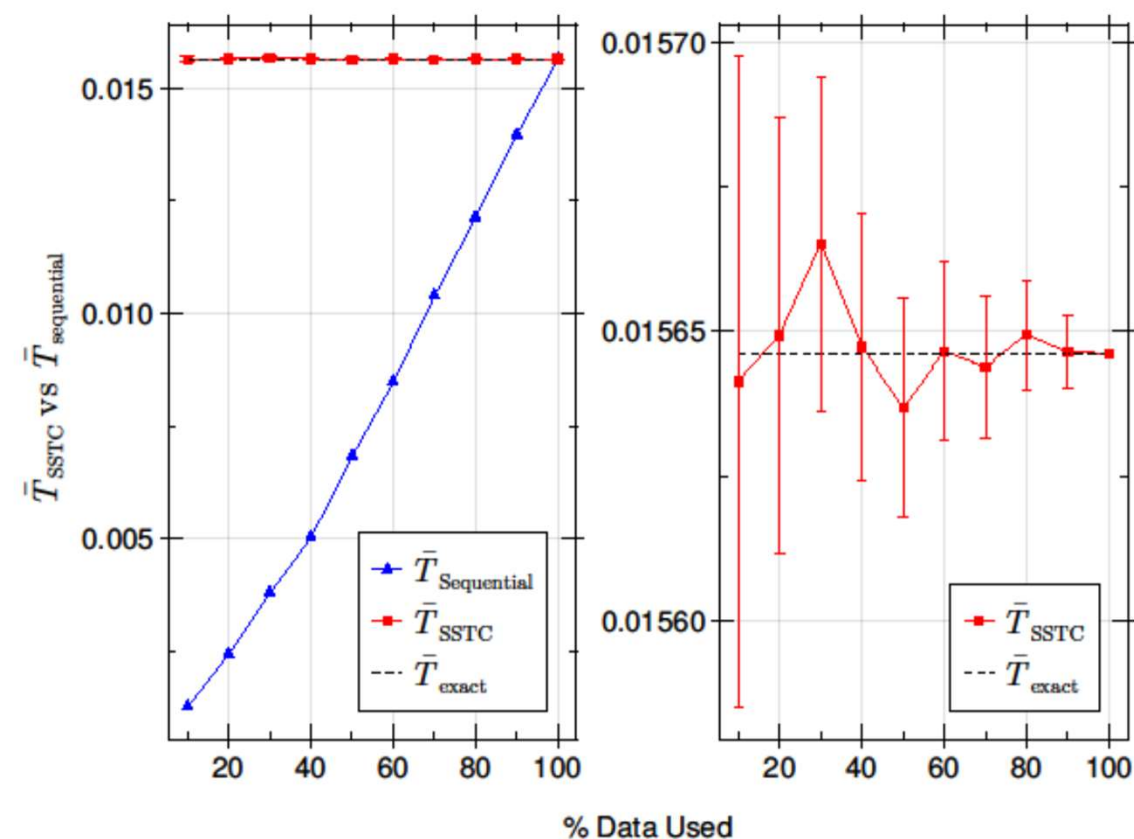
Results: Two-point Kernel

$$T_{\text{exact}} = \sum_{p_1 q_1 p_2 q_2}^N C_{p_1 q_1 p_2 q_2} \langle \phi_{p_1}(\mathbf{r}_1) \phi_{p_2}(\mathbf{r}_2) | r_{12}^{-1} | \phi_{q_1}(\mathbf{r}_1) \phi_{q_2}(\mathbf{r}_2) \rangle$$



Results: Four-point Kernel

$$T_{\text{exact}} = \sum_{p_1 q_1 p_2 q_2 p_3 q_3 p_4 q_4}^N C_{p_1 q_1 p_2 q_2 p_3 q_3 p_4 q_4} \langle \phi_{p_1}(\mathbf{r}_1) \phi_{p_2}(\mathbf{r}_2) \phi_{p_3}(\mathbf{r}_3) \phi_{p_4}(\mathbf{r}_4) | g(r_{13}) r_{12}^{-1} g(r_{24}) | \phi_{q_1}(\mathbf{r}_1) \phi_{q_2}(\mathbf{r}_2) \phi_{q_3}(\mathbf{r}_3) \phi_{q_4}(\mathbf{r}_4) \rangle$$



Results: 1-particle Green's Function

Ionization Potentials of PbS quantum dots

Iterative Solution: $\omega_0 + \Sigma(\omega) = \omega$

System	Koopmans (eV)	SSTC-IP (eV)	Error in IP (eV)
Pb ₄ S ₄	8.28	7.66	0.01
Pb ₄₄ S ₄₄	7.13	6.96	0.01
Pb ₁₄₀ S ₁₄₀	6.91	6.71	0.08

Key Points:

- The IPs of the PbS QDs decreased with increasing dot size
- Error in IP is two orders of magnitude smaller than IP

Nicole Spanedda, Peter F. McLaughlin, Jessica J. Beyer, and Arindam Chakraborty, "Investigation of ionization potential in quantum dots using stochastically enumerated stratified sampling of molecular orbitals method" (JCTC 2023)

Results: Exciton Binding Energies

Multicomponent Coupled Cluster calculation using SSTC

Exciton binding energy in CdSe quantum dots

All energies are in eV

Dot diameter (nm)	System	SSTC-mcCC-SD ¹	Sequential mcCC-SD ²
<1	Cd ₆ Se ₆	3.33	3.33
2	Cd ₈₃ Se ₈₁	0.56	0.60
3	Cd ₂₃₂ Se ₂₃₅	0.36	0.36
4	Cd ₅₃₄ Se ₅₂₇	0.29	0.28

▪ **Statistical error in Stochastic mcCC:** less than 0.01 eV

¹ Martin, Chakraborty, to be submitted (**this work**)

² Ellis, Chakraborty JCTC 2020

Results: Biexciton Formation Energies

Multicomponent Coupled Cluster calculation using SSTC

All energies are in eV

Cluster	SSTC-mcCC-SD	Sequential mcCC-SD
$\text{Ga}_{140}\text{As}_{140}$	1.57	1.58
$\text{Pb}_{140}\text{S}_{140}$	3.30	3.31
$\text{Cd}_{144}\text{Se}_{144}$	1.46	1.44

- Statistical error in Stochastic mcCC: less than 0.01 eV

1 Martin, Chakraborty, to be submitted (**this work**)

Summary

$$\sum_{abcij} \approx \sum_{\mathbf{k}} \text{⌘}$$

- Stratified Stochastic Tensor Contraction (SSTC) provide an efficient approach for performing large-scale tensor contraction
- Statistical error in the sampling procedure can be controlled by using stratification
- Sampling without replace (swr) is a form of correlated sampling and is a crucial part of the algorithm